Assignment 10, due October 19

For this assignment, read in the weekly Billboard Top 100 songs data covering the years up to 2021 from `http://manctsui.github.io/datasets/charts.csv` into SAS as the dataset called "charts".

Problem 1

Write SAS code that only returns the table consisting of the top 10 ranking songs from each week, and the columns returned are: date, rank, song, artist. Store this as a dataset called "topten".

Problem 2

Return only those rows of "topten" where the song is by Ed Sheeran or by Olivia Rodrigo. Store this as a new "ed_or_olivia"

Problem 3

Create two extra columns to "charts" called "topTenThisWeek" and "topTenLastWeek" which are true or false according to whether the song is in top 10 this week or last week, respectively.

Problem 4

Create a 2-way frequency table that determines how many songs are

- top 10 this week and top 10 last week
- top 10 this week but not top 10 last week
- not top 10 this week but top 10 last week
- not top 10 this week and not top 10 last week

(Choose to deal with missing data however you wish.)

Note: SAS does not immediately understand a column named `last-week` cannot be directly because of the dash, so you must refer to it as a name literal like `"last-week"n`.

You already saw an example of a hexadecimal literal which uses the literal suffix "x" instead: the tab character is the 9th character in ASCII and so we can refer to it as `'09'x` in SAS.